

are dynamic places where important landscape-altering events take place, such as volcanic activity, mountain-building, island formation, rifting, and earthquakes.

Plate boundaries fall into three main classes. The first, called divergent boundaries, occur where two plates are moving apart. At this type of boundary, material continuously welling up from the mantle fills the gap and creates new plate. Divergent boundaries are found extensively on the ocean floors, in the form of mid-ocean spreading ridges. But they also occur, among other places, in Iceland and in East Africa, along what is called the Great Rift Valley (see pp.184–87). They are always associated with volcanic activity.

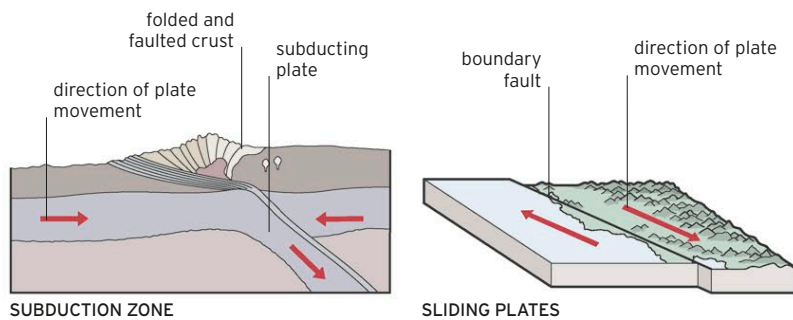
The second class of boundaries, convergent boundaries, are found where two plates move toward each other. At these boundaries, all or some of one plate moves down, or subducts, under the other and is destroyed. Where both plates carry continental crust, mountains form as chunks of crust are forced together. This process accounts for the origin of many mountain chains, such as the Himalayas. Other instances involve a plate carrying oceanic crust subducting the other plate. Boundaries of this type are characterized by features such as a deep-sea trench along the line of the boundary; a chain of volcanoes, always on the side of the plate that is not subducting; and the frequent occurrence of major earthquakes.

The third class of boundaries, called transform boundaries, occur where plate edges push past each other, without new plate being created or existing plate destroyed. These boundaries are also a site and source of earthquakes. Examples can be found in California (along the famous San Andreas Fault, see pp.30–31), in New Zealand's South Island, and elsewhere, including extensively on the ocean floors.

Hotspot volcanism

Although two of the three main types of plate boundary are frequently linked to volcanism, not all volcanoes develop at plate boundaries. Some appear in the middle of plates. This type of volcanism is usually accounted for by the presence of mantle hotspots—locations at the top of Earth's mantle that appear to be the source of peculiarly large amounts of energy. The movement of a plate across a hotspot (whose position is fixed) can, over a long period of time, create a chain of volcanic features at the surface. This explains, for example, evidence of much ancient volcanic activity in a line to the southwest of Yellowstone National Park (due to movement of the North American Plate over this hotspot). It is also regarded as the best explanation for the formation of some chains of volcanic islands, including the Hawaiian Islands (see pp.316–19).

Plates move at a rate of between $\frac{1}{4}$ in (7 mm) per year—one-fifth the rate that fingernails grow—and **6 in (150 mm) per year**—the rate human hair grows



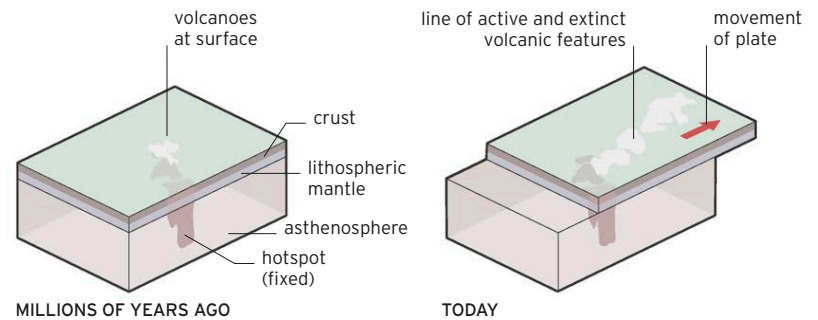
△ CONVERGENT BOUNDARY

At this type of boundary, one plate moves under, or subducts, another and is gradually destroyed (top). If both plates carry continental crust, mountains are pushed up (bottom).



△ TRANSFORM BOUNDARY

At a transform boundary, two plates push past each other in opposite directions (top). This type of boundary may be visible as a linear fault running across the landscape (bottom).



△ VOLCANIC CHAINS

Where a plate moves over a mantle hotspot, it can lead to a chain of volcanic features at the surface. The Hawaiian Islands formed in this way through movement of the Pacific Plate over a hotspot currently located under the volcano Kilauea (bottom) on the Big Island of Hawaii.